Theodore Halpern

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Education

Georgia Institute of Technology Atlanta, GA

Bachelor of Science in Computer Science | *GPA 3.67* *May 2027*

Relevant Coursework: Computer Networking, Operating Systems, Algorithms, Linear Algebra, Digital Design

Experience

Software Engineering Intern May 2026 – August 2026

Meta Platforms, Inc. *Seattle, WA*

- Reduced production query latency from 30+ seconds to 400ms and eliminated out-of-memory failures and timeouts by replacing in-memory filtering with indexed MySQL queries.
- Independently identified and removed 861 invalid Meta CLI flags, preventing thousands of monthly AI tool call failures.
- Discovered and repaired 500+ broken unit tests, then adapted application-based CI to run library tests, adding automated regression checks for the team's core dashboard library.
- Built AI agent skills that save investigation findings as chart annotations and reuse them to inform future investigations.

Software Engineering Intern January 2026 – April 2026

Uber Technologies, Inc. *San Francisco, CA*

- Built a pricing data pipeline ingesting PrestoDB trip data into DuckDB across 10+ dimensions; the service reduced anomaly-investigation queries from 25 minutes to 8 seconds through DuckDB aggregation and pandas-based detection.
- Shipped a React/FastAPI application enabling pricing teams to identify statistically significant anomalies, compare trendlines across data aggregates/filters, and pinpoint the components driving each anomaly.
- Created a Slack alerting pipeline with an AI root-cause agent that searches internal tickets, key tables, and code changes.

Software Engineering Intern May 2025 – August 2025

Meta Platforms, Inc. *Seattle, WA*

- Architected streaming from a GraphQL Hack (PHP) server to the React Native app, reducing Facebook Marketplace feed load latency by 50ms.
- Refactored server-side Marketplace listing generation for incremental delivery, enabling early client-side rendering.
- Validated product improvements by designing and executing A/B tests that identified statistically significant performance gains between control and treatment groups.

Data Structures and Algorithms Teaching Assistant January 2025 – Present

Georgia Tech College of Computing *Atlanta, GA*

- Facilitated office hour sessions, offering personalized guidance to students on advanced topics, including Dynamic Programming, Rabin–Karp pattern matching, and Binary Search Trees.
- Graded homework and exams, providing individualized feedback on algorithmic time complexity and Java coding style.

Software Engineering Intern - Meta University June 2024 – August 2024

Meta Platforms, Inc. *Menlo Park, CA*

- Built IndoorMaps, enabling users to draw floor plans, search rooms, share maps, and invite other editors.
- Implemented Dijkstra indoor routing with visibility graphs; cached graphs in Redis and invalidated them after map edits.
- Built a WebSocket server directly on Node.js TCP sockets, implementing HTTP upgrades, frame parsing, and JWT authentication for live location sharing.

Projects

BikeBuddy | *C++, ESP32, & SPI* | 1st Place Hardware Track at HackGT 12 September 2025

- Integrated ESP32 cameras and TFT SPI screens; resolved brownouts through circuit debugging and power optimization.
- Developed 3-way WiFi AP and client communication between React Native, the rear ESP32-CAM, and the front ESP32 C6.

Technical Skills

Languages: TypeScript, JavaScript, Java, C++, C, Python, SQL, Hack/PHP, Go, VHDL
Libraries: React, React Native, Next.js, Node.js, FastAPI, GraphQL, Apollo, Relay
Databases: PostgreSQL, MySQL, Redis, DuckDB, PrestoDB
Tools: Git, Docker, ESP-IDF, Claude Code, Codex